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ABSTRACT

High-purity single-crystal wide-bandgap semiconductor cubic silicon carbide (3C–SiC) has the second-highest thermal conductivity among wafer-scale crystals (after diamond), making it ideal for thermal management in electronic devices. However, doping—essential for electrical property tuning—may significantly affect its thermal conductivity. While numerous theoretical studies exist, experimental data remain limited. In this work, the thermal conductivity of heavily nitrogen-doped 3C–SiC single crystals grown via the top-seeded solution growth method is measured by time-domain thermoreflectance. The results show a significant reduction (up to 30%) in thermal conductivity at nitrogen doping concentrations around 2×10^{20} cm^{-3}. The doping concentration and distribution are investigated using secondary ion mass spectroscopy and atom probe tomography, revealing an atomic-scale uniform nitrogen distribution. Experimental results show a lower thermal conductivity reduction than previous density functional theory predictions, indicating weaker phonon–electron scattering than expected. Large-area thermal conductivity measurement and mapping reveal spatially uniform thermal conductivity in 3C–SiC at the micro- to-macroscale, emphasizing its practical utility and general high quality. These findings shed light on understanding the doping effects on thermal transport in semiconductors and support further exploration of 3C–SiC for electronics thermal management.

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I. INTRODUCTION

SiC is a promising wide-bandgap semiconductor material possessing a high breakdown electric field, high thermal conductivity, and high electron mobility.[1–3] These properties position SiC as a potential leader in revolutionizing high-power devices and as a promising thermal management material.[4–8] Among the three commonly used polytypes of SiC—4H–SiC, 6H–SiC, and 3C–SiC—the cubic-phase 3C–SiC polytype stands out due to its simpler crystal structure, which is predicted to exhibit the highest thermal conductivity.[9,10] Additionally, 3C–SiC possesses the highest electron saturation velocity among these polytypes, which further enhances its attractiveness for energy-efficient power switching devices.[2]

3C–SiC is also the only polytype that has been successfully grown directly on a silicon substrate, enabling direct integration with silicon-based logic device.[9] As overheating becomes an increasingly critical issue in electronic devices, particularly in highly integrated silicon-based systems like 3D integrated circuits, where the power density is extremely high and thermal dissipation is constrained,[11–16] the superior thermal performance and compatibility of 3C–SiC could provide a significant solution. High-concentration nitrogen (N) doping is commonly used in the electronics industry to produce *n*-type SiC.[17–19] However, introducing impurities inevitably reduces thermal conductivity due to increased defect–phonon scattering.[20,21] Beyond conventional mass-induced Rayleigh scattering, strong phonon–electron scattering caused by the high electron density excited by heavy nitrogen doping may lead to additional thermal

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conductivity degradation. This effect can further degrade thermal conductivity beyond the impact of mass doping alone,[10] posing a challenge for thermal management in electronic devices. When N impurities are lightly doped, the impact of electron–phonon scattering on thermal conductivity is generally negligible. Previous experimental and theoretical studies support this view for 4H–SiC and 6H–SiC under light N-doping conditions,[10,22] However, under heavy N-doping situation, the theoretical work of Pang *et al.*[23] suggested that phonons can be strongly scattered by electrons excited from nitrogen donors, leading to an additional reduction in lattice thermal conductivity. Wang *et al.*[24] also studied the effect of electron–phonon scatterings in 3C–SiC. However, their research was limited to a pure electron or hole-doping condition and did not consider the phonon–defect scattering effect of elemental dopants (such as N dopants) themselves. Similar theories and experimental findings in silicon have been reported by Liao *et al.*[25,26] and Dongre *et al.*,[27] who studied the coexistence of defect–phonon and electron–phonon scatterings in heavily doped Si. Despite the fact that phonon scattering mechanism is crucial for the fundamental understanding and a broader application of 3C–SiC materials, it remains to be experimentally and systematically verified due to the limitations in obtaining high-quality 3C–SiC single crystals and advanced thermal/dopant characterization techniques.

Unlike 4H–SiC with its mature growth techniques, the lower growth rate and inferior crystal quality of 3C–SiC have hindered further research on this material.[28,29] Recently, the top-seeded solution growth (TSSG) method for 3C–SiC has been developed, enabling the growth of high-quality single-crystal wafers with high N-doping concentration up to $1.89 \times 10^{20}\,\mathrm{cm}^{-3}$.[30] This breakthrough facilitates simultaneous achievement of high crystal quality and substantial N doping, providing insights into the underlying principles of phonon scattering in heavily doped 3C–SiC. Understanding the mechanisms of thermal transport in this system, as well as the quantitative effects of doping, is crucial for optimizing growth conditions to balance electrical and thermal performance.

In this study, heavily N-doped 3C–SiC single crystals were grown on 4H–SiC substrates via the TSSG method.[30] The crystal structure and quality were confirmed by Raman spectroscopy, while thermal conductivity was measured using time-domain thermoreflectance (TDTR).[31] Additionally, secondary ion mass spectroscopy (SIMS) and atom probe tomography (APT) were employed to quantify the doping concentrations and atomic spatial distribution of different doping elements. To better understand the macroscale effect of dopant distribution and crystal quality, large-area thermal conductivity testing and mapping were performed, providing a visual representation of its impact. The Callaway[32] model was used to extract the effect of Al impurity on thermal conductivity using Matthiessen's rule.[33]

II. METHODS

Three heavily N-doped 3C–SiC samples, labeled sample A, sample B, and sample C, were investigated in this work. These high-quality single crystals (3C–SiC) were grown by crystallizing C and Si atoms from the melt in a crucible. The Si-based solvents are loaded into a high-purity graphite crucible, which served as both the reaction container and carbon source. An axial temperature gradient from the bottom to top is set to be 5–15 °C cm^{-1} with the lowest temperature of ~1850 °C at the top surface. The 4 in. 4H–SiC was used as the seed crystal with the (0001) surface oriented as the growth front. More experimental details and crystal quality validation can be found in Ref. 30. The high crystal quality was confirmed by a combination of Raman spectroscopy, atomic-resolution STEM, and XRD. The stacking fault (SF) density was discovered to be record-low, and the average full width at half maximum (FWHM) of x-ray rocking curve was measured to be 30 arcsec.[30] These results collectively indicate that the influence of crystal defects, such as SFs and grain boundaries, is negligible. Therefore, the thermal conductivity of ~500 W/m K measured for defect-free 3C–SiC in the prior work serves as a valid and sufficient benchmark for our study.[9] Further details can be found in Fig. S1 in the supplementary material (SF density comparison) and Fig. S2 in the supplementary material [x-ray curve (XRC) FWHMs]. The samples tested in this study were grown under N_2 pressures of 25–30 kPa to optimize the growth condition of 3C–SiC over 4H–SiC phase and to achieve a high concentration of N doping. The as-grown 3C–SiC ingots are then cut, ground, and polished for the following characterizations.

Raman spectroscopy was used to verify the structure of the samples. The measurements were performed using a Horiba LabRAM confocal Raman imaging system, equipped with a 50× objective lens. Spectra were acquired using 5% of the total laser power at a wavelength of 532 nm, with an acquisition time of 60 s over five accumulations. The peaks in the Raman spectra corresponded to the energies of different phonon branches, confirming the crystal structure. Both the frontside (3C–SiC film) and the backside (4H–SiC substrate) of the sample were characterized by Raman. Further measurement details are provided in the supplementary material, with the Raman spectra acquired from the 4C–SiC substrate shown in Fig. S3 in the supplementary material.

SIMS was used to determine the concentrations of N and other melt metals. The SIMS analysis was conducted by Evans Material Technology (Shanghai) Co., Ltd., with equipment provided by Cameca. Cs^+ ions were accelerated to 10 keV to detect the N element, and O_2^+ ions were accelerated to 12.5 keV to detect the Al element. To further study the dopant special distribution in the material, excluding the effect of non-uniform doping, APT was performed to provide the elemental distribution of N with atomic resolution.[34] A needle shaped sample used for APT characterization was fabricated by focus ion beam (FIB) milling with a FEI Helios G5 DualBeam system. A 50 nm platinum protection layer was coated onto the sample's surface. The test was conducted by a LEAP 5000 XR from Cameca with potential charged between 5 and 9 kV. 355 nm laser pulses of 30 pJ at a frequency of 200 kHz were used to evaporate the surface atoms, under the condition of 40 K and 1×10^{-11} Torr. Cameca's AP Suite 6.3 code was used to reconstruct the 3D distribution of elements Si, C, N, and Al. The APT characterization was performed at the Beijing University of Science and Technology Analysis and Testing Center Co., Ltd.

The thermal conductivity of the three samples was measured by TDTR, an optical thermal characterization method based on an ultrafast femtosecond laser, with schematic diagram shown in Fig. S1 in the supplementary material. A sapphire: Ti (Mai Tai)

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laser source, operating at 80 MHz with a wavelength of 785 nm and a pulse width shorter than 100 fs, generated a train of laser pulses, which was then split into a probe beam and a pump beam. The pump beam was modulated at a specific modulation frequency controlled by a signal from a function generator. The pump and probe beams were then recombined, and an objective lens focused them onto the sample surface. A mechanical delay stage was used to adjust the arrival time of the pump beam relative to the probe beam. The pump beam induced periodic temperature variations on the sample's surface, and these thermal signals were detected by the probe beam via thermoreflectance. A ~ 80 nm thick Al layer was deposited on the sample surface as a transducer to enhance the signal. The reflected beam was captured using a photodiode and an RF lock-in amplifier. By fitting the experimental data to a theoretical heat transfer solution of the multilayer sample, the unknown thermal properties were obtained. In this study, a modulation frequency of 10.1 MHz and a spot radius of 12 μm were chosen to avoid any possible ballistic thermal transport phenomenon.[9,35,36] Further theoretical and experimental details on TDTR measurements can be found in Refs. 31 and 37–40.

Large-area thermal conductivity mapping was performed by a two-color TDTR system[41] with a pump and probe beam root mean square (RMS) radius of about 10 μm at a modulation frequency of 5.02 MHz to validate the uniform high quality of the sample. A piezoelectric 2D positioning stage (Physik Instrumente, Q-545.240) was used to control the micrometer-scale movement of the sample. The delay stage was fixed at a specific position so that the sensitivity of Al/SiC thermal boundary conductance (TBC) is negligible, ensuring that the ratio signal R solely reflected the thermal conductivity of 3C–SiC. More details about the principle of TDTR mapping and the sensitivity analysis of parameters can be found in Ref. 42–45 and Secs. II–IV of the supplementary material (with the experimental setup of TDTR shown in Fig. S5 in the supplementary material).

The samples in this work were unintentionally doped with Al. To exclude the effect of the Al impurity, the Callaway model for thermal conductivity was employed. For semiconductors such as 3C–SiC, the heat is carried by phonons instead of electrons, and the thermal conductivity can be expressed as[46]

$$\kappa = \frac{1}{3} C_v v_a^2 \tau, \tag{1}$$

where κ is the thermal conductivity, C_v is the volumetric heat capacity of the material, v_a is the average phonon group velocity, and τ is the phonon relaxation time. The average phonon group velocity v_a is given by

$$v_a^3 = \frac{1}{3}\left(\frac{1}{v_l^3} + \frac{2}{v_t^3}\right), \tag{2}$$

where v_l is the longitudinal sound velocity and v_t is the transverse sound velocity.[47,48] The relaxation time (τ) incorporates contributions from various scattering mechanisms, including boundary, impurity, electron, and phonon–phonon scattering. According to Matthiessen's rule, the total relaxation time τ is related to the individual relaxation time induced by each scattering source,[33,49,50]

$$\frac{1}{\tau} = \frac{1}{\tau_{pb}} + \frac{1}{\tau_{pp}} + \frac{1}{\tau_{pe}} + \sum_i \frac{1}{\tau_{pd_i}}, \tag{3}$$

where τ_{pb}, τ_{pp}, τ_{pe}, and τ_{pd} denote the relaxation times for phonon–boundary, phonon–phonon, phonon–electron, and phonon–defect scattering, respectively. The total phonon–defect scattering relaxation time τ_{pd} is the integrated effect of all kinds of defects τ_{pdi}. The Al impurity-induced scattering rate was extracted from Katra *et al.*[51] by subtracting the undoped sample's scattering rate from the Al-doped sample's total scattering rate; the details are discussed in Sec. V of the supplementary material. This contribution was then subtracted from the total scattering rate via Eq. (3) to isolate the effects of N dopants.

III. RESULTS AND DISCUSSION

The concentrations of different dopants revealed by SIMS are listed in Table I. N is the dominant impurity in these samples with concentrations approximately two orders of magnitude higher than that of Al. However, according to the previous theoretical study,[51] the low Al concentration reduces the thermal conductivity slightly. Therefore, the influence of Al dopants was excluded by the method mentioned in Sec. II. Given the similar doping concentrations and growth conditions across all three samples, their dopant distribution trends were assumed to be identical.

The Raman spectrum of the sample, shown in Fig. 1(a), exhibits a distinctive Raman shift peak at 794 cm^{-1}, corresponding to the transverse optical (TO) phonon mode at the Γ point of 3C–SiC. The theoretical phonon dispersion relation for 3C–SiC, highlighting different branches, is depicted in Fig. 1(b). The phonon energy corresponding to the Raman peak obtained in this study is included for comparison and agrees well with reference values,[52–54] confirming the cubic-phase structure and high quality of the sample. The atomic-scale distribution of the major elements, as obtained by APT, is shown in Figs. 1(c)–1(e). Figs. 1(c) and 1(d) demonstrated the spatial distribution of silicon and carbon atoms, respectively. Both elements, being the primary constituents of the SiC host material, exhibit uniform and dense distributions. The atomic distribution map of N dopants is shown in Fig. 1(e). To prevent ambiguity between Si^{2+} and N^+ ions—which exhibit nearly identical charge-to-mass ratios—we reconstructed the N-dopant distribution using N^{2+} ions. This approach ensures accurate spatial mapping while preserving the representative uniformity of N-dopant incorporation. Figure 1(e) reveals a uniform distribution of N dopants in atomic-to-nanoscale. This spatial homogeneity excludes the

TABLE I. Doping concentrations of three 3C–SiC samples measured SIMS.

Samples	A	B	C
N concentrations (atoms cm^{-3})	2.29×10^{20}	2.49×10^{20}	2.62×10^{20}
Al concentrations (atoms cm^{-3})	5.65×10^{18}	2.12×10^{18}	8.27×10^{18}

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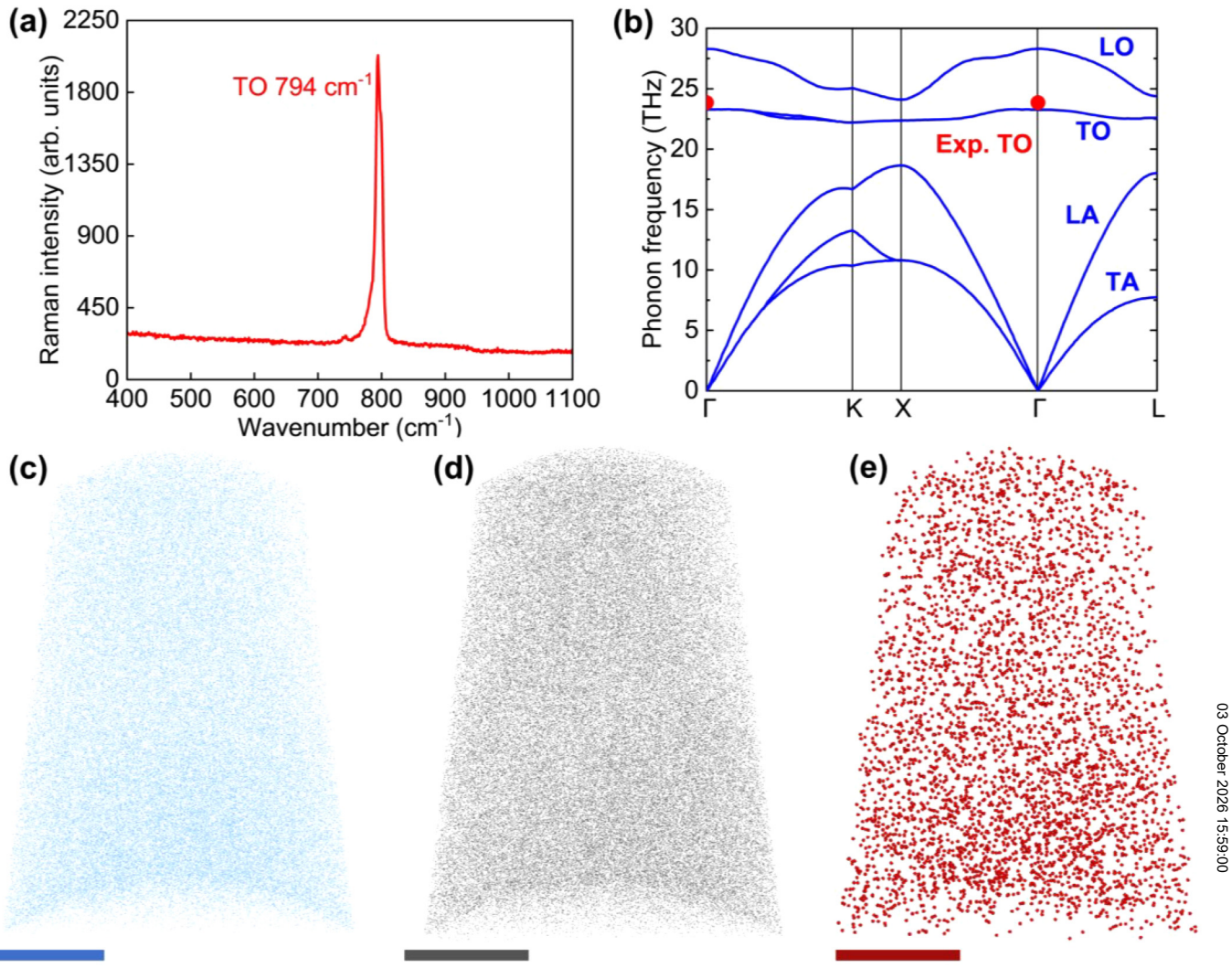


FIG. 1. Structural characterization of the 3C–SiC sample. (a) Raman spectrum of the 3C–SiC sample from this study, with the peaks corresponding to different phonon modes. Only the peak of TO phonon branch can be observed, which is similar to Ref. 30. (b) The phonon dispersion relation of 3C–SiC, where the blue curve represents the theoretical results from Ref. 24. Different phonon branches are labeled LO (longitudinal optical), TO (transverse optical), LA (longitudinal acoustic), and TA (transverse acoustic). The corresponding Raman peak of this sample is depicted in a red circle. (c)–(e) Elemental distribution maps in atomic resolution revealed by APT: (c) Si, (d) C, (e) N. The scale bar is 20 nm.

potential of phonon–defect scattering center decrease.[55] We attempted to map the distribution of unintentionally doped Al atoms. However, due to the low concentration of Al dopants, almost no data counts were obtained, as shown in Fig. S4(a) in the supplementary material. For comparison, Fig. S4(b) in the supplementary material presents the simultaneous distribution of all four elements.

Figure 2 shows the measured thermal conductivity of samples A, B, and C, along with theoretical predictions based on different scattering mechanisms. In Fig. 2(a), the TDTR signal ratio is plotted as a function of delay time. The red solid line represents the best-fit curve for 3C–SiC thermal conductivity (κ), while the two blue dashed lines indicate calculated ratio curves corresponding to ±10% variations in κ. More fitting results can be found in Figs. S6–S8 in the supplementary material. The significant deviation between the red solid line and blue dashed lines demonstrates the high sensitivity of our κ measurements, and the detailed sensitivity plot of the two fitting parameters is shown in Fig. S9 in the supplementary material. By removing the Al scattering rates from the measured data by the Callaway model, the thermal conductivity values without Al impurities are obtained, as indicated by the blue open triangles in Fig. 2(b). The measured results are shown as red

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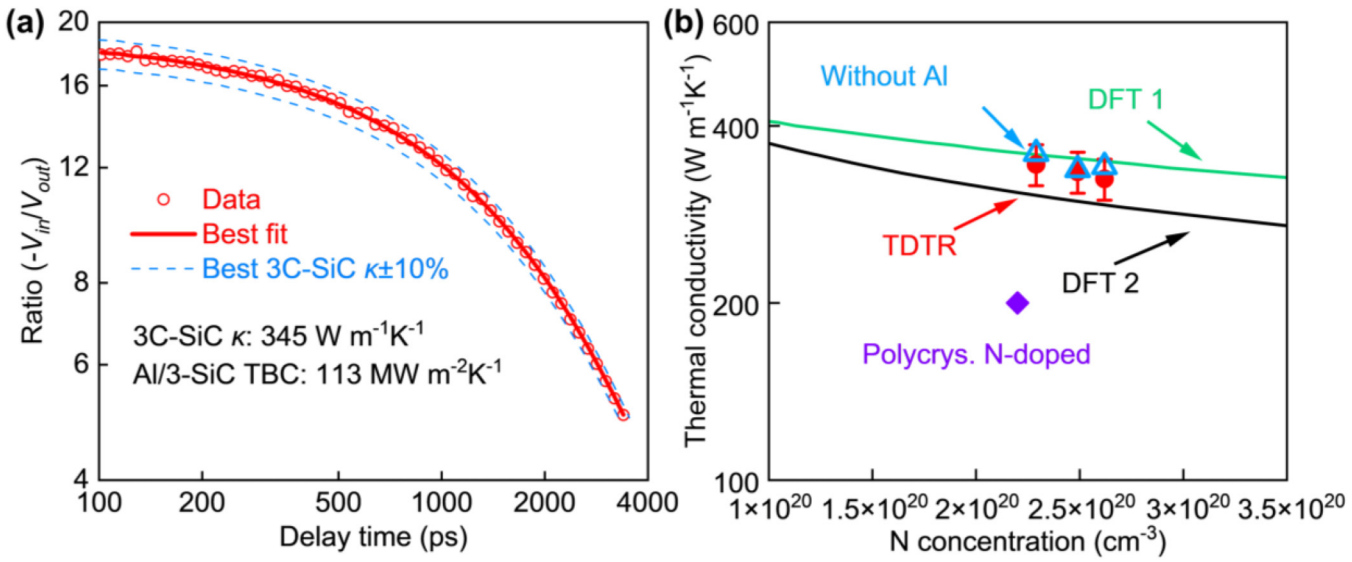


FIG. 2. Comparison of measured thermal conductivity for the three samples with previous experimental and theoretical results. (a) Typical TDTR signal ratio with the best-fit curve, along with the curve with ±10% best-fit 3C–SiC κ. (b) Experimental results from this work are shown as red circles, while the derived thermal conductivity values that exclude the influence of Al impurities are indicated by the blue triangles. The measured thermal conductivity of polycrystalline N-doped 3C–SiC from Ivanova *et al.*[56] and the DFT calculated results from Katre *et al.*[51] (marked as DFT 1) and Pang *et al.*[23] (marked as DFT 2) are also included. The error bar of the measurement is ±8%.

circles, which are slightly lower than the derived values due to the presence of unintentionally doped Al impurity. Previous experimental result of polycrystalline 3C–SiC ceramic from Ivanova *et al.*[56] was also included for comparison. These literature data show significant deviations from theoretical predictions, likely due to grain boundary-induced phonon scattering and doping concentration uncertainties. The theoretical thermal conductivity that only considers phonon–defect scattering is plotted as a green solid line,[51] and the theoretical thermal conductivity that considers both phonon–defect and phonon–electron scatterings is plotted as a black solid line[23] in Fig. 2(b). The results indicate that the influence of electron–phonon scattering due to N doping is less pronounced than predicted by Pang *et al.*[23] at a doping concentration of 10^{20} cm^{-3}. Specifically, when the N-doping concentration increases from 1×10^{20} to 2×10^{20} cm^{-3}, a significant reduction of 16% in κ is predicted for the strong electron–phonon scattering model. This is in contrast to the 9% reduction predicted by pure phonon–defect DFT calculations. However, according to the research of Pang *et al.*,[23] at such a high N-doping concentration of 2×10^{20} cm^{-3}, the heat transport in 3C–SiC is expected to become dominated by defect–phonon scattering. In contrast, electron–phonon scattering is predicted to remain significant at a lower concentration of $\sim 2 \times 10^{19}$ cm^{-3}. This is consistent with the prediction from Wang *et al.*, who reported a ~10% drop of κ for pure electron-doped 3C–SiC at an electron concentration of 1×10^{19} cm^{-3}.[24]

The macro-scale crystal quality and thermal conductivity of the sample—a critical factor for device fabrication—may exhibit spatial variations due to the high doping concentration. To assess the uniformity of doping and the overall crystal quality of the samples at micro-to-macroscale, we measured thermal conductivity at 15 random locations on the 3C–SiC samples. For each location, a complete standard TDTR measurement was performed to obtain both the 3C–SiC κ and Al/SiC TBC. The results, presented in Fig. 3(a), demonstrate good uniformity across the samples. The variation in thermal conductivity for each sample is less than 15%, with most data points fluctuating within approximately 10%, compared to the 8% systematic error bar of the TDTR measurement. This indicates that the thermal conductivity is uniformly distributed across the samples, supporting the conclusions discussed earlier and confirming the high and consistent quality of the 3C–SiC crystals. The larger deviations in TBC observed in sample C are attributed to variations in the Al/SiC interfaces and contamination of the 3C–SiC surface, which are unrelated to the underlying 3C–SiC crystals.

In addition, thermal conductivity mapping was performed on sample A to provide a direct visualization of the macro-scale thermal conductivity distribution. The sensitivity plot of three thermal parameters is shown in Fig. 3(b), and the fixed delay time was then chosen to be 280 ps to exclude Al/SiC interface effects, ensuring that the TDTR signal primarily reflects substrate thermal conductivity. The resulting thermal conductivity variation map of sample A, acquired over a $200 \times 200\,\mu m^2$ area with a 10 μm step size, is displayed in Fig. 3(c), which visualizes the uniform distribution of κ. Further statistical analysis of the conductivity variations is presented in Fig. 3(d), where the frequency distribution of percentage deviations is plotted. The data closely follow a fitted Gaussian distribution (red curve), with 92.5% of measurements falling within ±1.5% of the mean value and maximum deviations constrained to ±3%, confirming exceptional micro-to-macroscopic uniformity in the 3C–SiC crystal.

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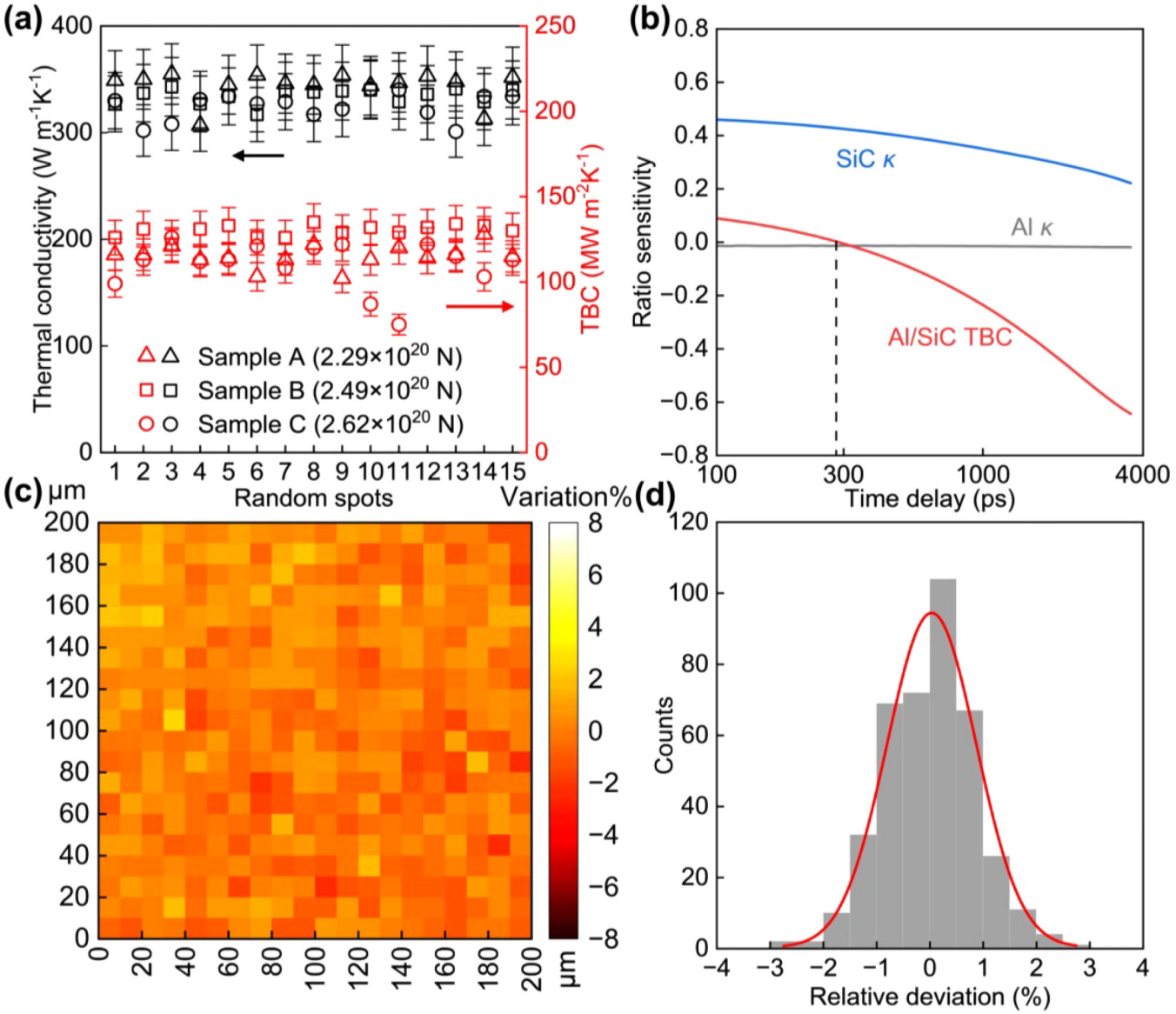


FIG. 3. The uniformity validation of the thermal conductivity of the N-doped 3C–SiC single crystal. (a) Measured thermal conductivity and Al/3C–SiC TBC at 15 random spots on each sample. Error bars indicate ±8% uncertainty. Black symbols represent the thermal conductivity, while red symbols represent the TBC. (b) Sensitivity of the Al/SiC TBC and the SiC thermal conductivity in TDTR. (c) 3C–SiC thermal conductivity variation map. (d) The frequency statistics histogram of different variation percentages and the corresponding best-fit Gaussian distribution curve.

To further investigate the thermal properties of heavily N-doped 3C–SiC, we performed TDTR measurements over a temperature range of 300–700 K, as shown in Fig. 4, along with previous experimental and theoretical results. Details about the temperature-variant TDTR experimental process can be found in Sec. II of the supplementary material. At high temperatures, phonon–phonon scattering dominates, while the defect–phonon scattering rate remains roughly constant, thereby diminishing the impact of impurities on thermal conductivity.[23] Due to the similar doping concentrations of N impurities, the three samples exhibit a minimal thermal conductivity variation. In logarithmic coordinates, the slope of each curve is approximately −1, indicating that at high temperatures, the κ of all three samples is proportional to T^{-1}, suggesting that phonon–phonon Umklapp scattering is the dominant process in this temperature range.[57]

The theoretical DFT prediction of the doping concentration of $1 \times 10^{20}/\text{cm}^3$ from the work of Pang *et al.* work is depicted as a

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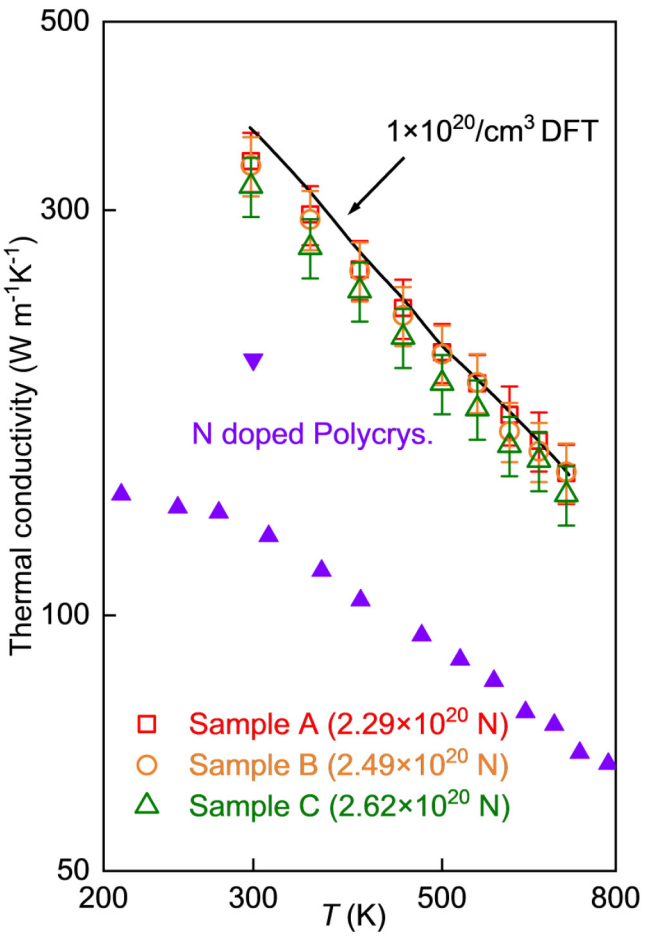


FIG. 4. Thermal conductivity of heavily N-doped 3C–SiC crystals at three different doping concentrations, measured over a temperature range of 300–700 K, compared with previous experimental data at various doping concentrations (represented by discrete triangles). The experimental results from Ivanova *et al.*[56] are also included, where the upright triangles correspond to a carrier concentration of 2.2×10^{20} cm^{-3}, while the inverse triangle corresponds to a carrier concentration of 8×10^{20} cm^{-3}.

black curve.[23] The measured thermal conductivity is lower than the predicted thermal conductivity near room temperature due to the higher concentrations of nitrogen, and they converge at high temperatures. This phenomenon can be attributed to two factors: first, the DFT calculation slightly overestimates the intensity of electron–phonon scattering in heavily N-doped 3C–SiC at room temperature, as mentioned earlier; second, at high temperatures, the predominant phonon scattering process is phonon–phonon Umklapp scattering with rate proportional to temperature, while the scattering rate of point–defect scattering is temperature independent.[58,59] Consequently, for the same material with identical intrinsic anharmonic scattering properties, thermal conductivity converges to the same value at sufficiently high temperatures, regardless of point–defect conditions.[60]

The experimental results for N-doped polycrystalline 3C–SiC ceramic from Ivanova *et al.*[56] are also included for comparison. The thermal conductivity measured in this study is much higher than the results of Ivanova *et al.*, as their samples are polycrystalline 3C–SiC with carbon mixed inside the samples, and the doping condition is inferred from carrier concentration, which is much lower than the actual doping concentration at room temperature. According to Pang *et al.*,[23] achieving a carrier concentration of the order of 10^{20} cm^{-3} at room temperature requires an N-doping concentration greater than 10^{21} cm^{-3}, an order of magnitude higher than the concentrations in this study. Despite the poor crystal quality, the trend of thermal conductivity at high temperatures follows the same linearity but with a smaller absolute slope of about 0.6 above room temperature. The reduced thermal conductivity is primarily attributed to the extensive grain boundary scattering caused by the polycrystalline structure and the strong phonon–defect scattering due to high doping concentrations. Even at temperatures as high as 1000 K, the more intense phonon–phonon Umklapp scattering does not outweigh the effects of these two factors.

IV. CONCLUSION

In summary, this study presents a comprehensive characterization of heavily nitrogen-doped 3C–SiC single crystals grown by the TSSG method across various N-doping concentrations. Raman spectroscopy confirmed the cubic crystal structure and high crystalline quality of the samples. SIMS and APT analyses revealed the dopant concentrations and spatial distributions, demonstrating atomic-scale uniformity in nitrogen distribution. Temperature-dependent thermal conductivity measurements (300–700 K) via TDTR were performed, with the influence of unintentional Al doping systematically eliminated using the Callaway model. Our results show that the additional electron–phonon scattering is less extensive than previously expected under a heavily N-doped condition with concentration as high as 2×10^{20} cm^{-3}. However, it remains possible that electron–phonon scattering plays a more significant role at a doping concentration of $\sim 1 \times 10^{19}$ cm^{-3}. Macro-scale multi-point thermal measurements and micro-scale area thermal conductivity mapping confirmed excellent sample homogeneity, with thermal conductivity variations primarily below 10%, comparable to the systemic error of TDTR (8%). These findings provide critical insights for optimizing 3C–SiC applications in both high-performance electronic devices and advanced thermal management systems.

SUPPLEMENTARY MATERIAL

See the supplementary material for additional details of the sample structural properties, TDTR experimental setup, TDTR measurement results, TDTR sensitivity analysis, and the detailed procedure on the removal of the impact of Al impurity.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Zifeng Huang: Data curation (lead); Investigation (equal); Validation (lead); Visualization (lead); Writing – original draft (lead); Writing – review & editing (equal). **Xufei Guo:** Methodology (equal); Software (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Yunfan Yang:** Investigation (equal); Resources (equal); Writing – original draft (supporting); Writing – review & editing (supporting). **Da Sheng:** Resources (supporting); Writing – original draft (supporting); Writing – review & editing (supporting). **Hui Li:** Conceptualization (equal); Funding acquisition (equal); Resources (equal); Supervision (equal); Writing – original draft (equal); Writing – review & editing (equal). **Yuxiang Wang:** Writing – original draft (supporting); Writing – review & editing (supporting). **Zixuan Sun:** Project administration (supporting); Supervision (supporting); Writing – original draft (supporting); Writing – review & editing (supporting). **Ming Li:** Writing – original draft (equal); Writing – review & editing (equal). **Runsheng Wang:** Supervision (supporting); Writing – original draft (supporting); Writing – review & editing (supporting). **Zhe Cheng:** Conceptualization (equal); Funding acquisition (equal); Methodology (equal); Project administration (equal); Supervision (equal); Writing – original draft (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding authors upon reasonable request.

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